Customer Shopping Behavior Analysis

1. Project Overview

This project examines customer shopping behavior using transactional data from **3,900 purchases** across multiple product categories. The objective is to analyze spending trends, customer segmentation, product performance, and subscription patterns to support informed business decision-making.

2. Dataset Summary

- **Rows:** 3,900
- **Columns:** 18
- **Key Attributes:**
  - **Customer Demographics:** Age, Gender, Location, Subscription Status
  - **Purchase Information:** Item Purchased, Category, Purchase Amount, Season, Size, Color
  - **Shopping Behavior:** Discount Applied, Promo Code Used, Previous Purchases, Purchase Frequency, Review Rating, Shipping Type
- **Missing Values:** 37 missing entries in the *Review Rating* column

3. Exploratory Data Analysis using Python

Data preparation and analysis were performed using Python.

• Data Loading

Imported the dataset using **pandas** for structured data handling.

• Initial Exploration

Used df.info() to examine data types and structure, and df.describe() to review summary statistics

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

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• Handling Missing Values
Identified null values and filled missing entries in the *Review Rating* column using the **median rating of the respective product category** to maintain consistency.

• Column Formatting
Renamed columns into **snake_case** format for improved readability and database compatibility.

• Feature Engineering
- Created an age_group column by categorizing customer ages into segments.
- Generated a purchase_frequency_days feature based on transaction data.

• Data Validation
Checked for redundancy between discount_applied and promo_code_used columns and removed unnecessary duplication to optimize the dataset.

• Database Integration
Connected the cleaned dataset to **PostgreSQL** and exported the structured DataFrame for SQL-based analysis.

4. Data Analysis using SQL
The following analyses were performed in PostgreSQL to answer key business questions:

1. Revenue by Gender
Calculated total revenue generated by male and female customers to compare spending behavior.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

2. High-Spending Discount Users
Identified customers who applied discounts yet made purchases above the overall average transaction value.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

Total rows: 839 Query complete 00:00:

3. Top 5 Products by Rating

Determined the products with the highest average customer review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. Shipping Type Comparison

Compared average purchase values between **Standard** and **Express** shipping methods.

	shipping_type text	round numeric
1	Standard	58.46
2	Express	60.48

5. Subscribers vs. Non-Subscribers

Analyzed differences in total customers, average spending, and total revenue based on subscription status.

	subscription_status text	total_customers bigint	avg_spend numeric	total_revenue numeric
1	Yes	1053	59.49	62645.00
2	No	2847	59.87	170436.00

6. Discount-Sensitive Products

Identified the top 5 products with the highest proportion of discounted purchases.

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.66
3	Coat	49.07
4	Sweater	48.17
5	Pants	47.37

7. Customer Segmentation

Classified customers into **New**, **Returning**, and **Loyal** categories based on their purchase history.

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

8. Top 3 Products per Category

Ranked the most purchased products within each product category.

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessories	Jewelry	171
2	2	Accessories	Sunglasses	161
3	3	Accessories	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions

Evaluated whether customers with more than five purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

10. Revenue by Age Group

Calculated total revenue contribution across different age groups.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

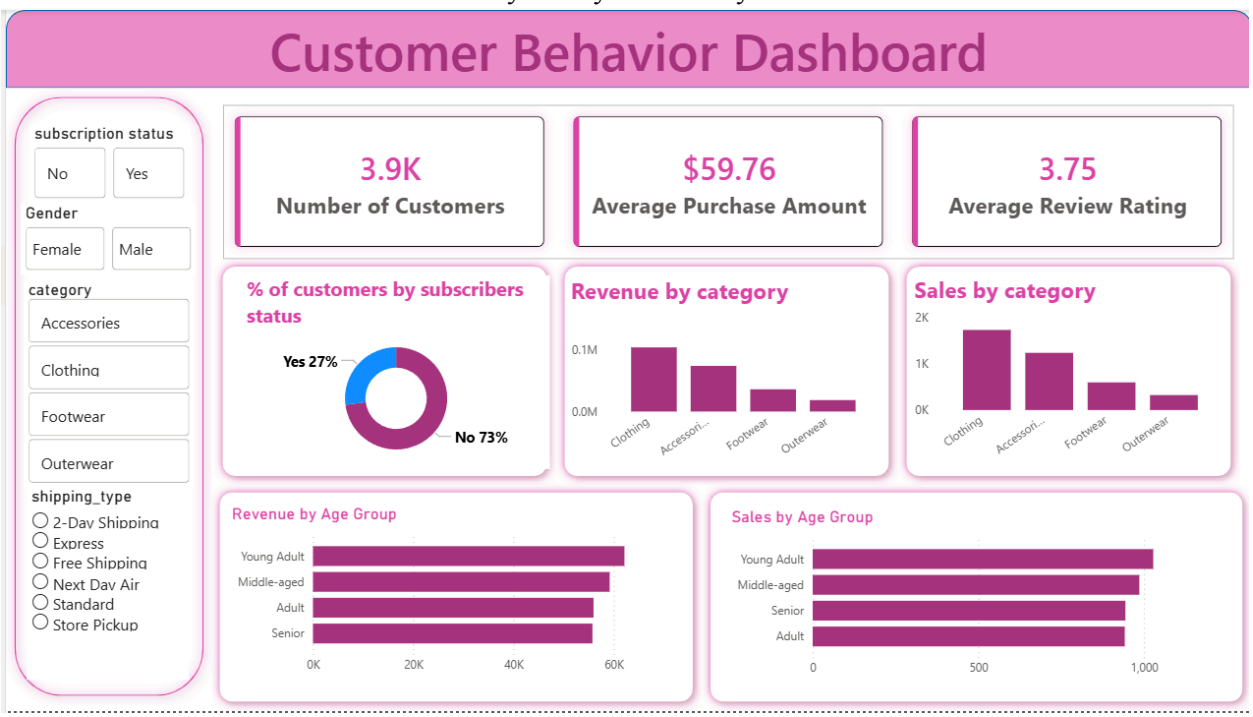
5. Dashboard in Power BI

An interactive dashboard was developed in **Power BI** to visually present the insights derived from the analysis.

The dashboard includes:

- Total Customers
- Average Purchase Amount
- Average Review Rating
- Revenue by Category
- Sales by Age Group
- Subscription Distribution

Filters were added for enhanced interactivity and dynamic analysis.



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6. Business Recommendations

- **Increase Subscription Engagement** – Offer exclusive incentives to encourage subscription growth.
- **Strengthen Loyalty Programs** – Reward repeat customers to move them into the loyal segment.
- **Refine Discount Strategy** – Optimize discounts to maintain profitability while driving sales.
- **Promote High-Performing Products** – Highlight top-rated and best-selling items in marketing campaigns.
- **Target High-Value Segments** – Focus marketing efforts on high-revenue age groups and premium shipping users.